

Muhammad Usama

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📧 MuhammadUsama100 🌐 muhammadusama100 📁 Portfolio

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INTRODUCTION

Research Assistant with 2 years of experience in multimodal generation, parametric shape reconstruction, and 3D retrieval/classification, plus 3 years of software engineering experience spanning distributed training, onprem/cloud deployment, and CI/CD. I bridge research innovation with robust, production-ready systems.

PUBLICATIONS

* equally contributing first author

- **NURBGen: Muhammad Usama***, Mohammad Sadil Khan*, Didier Stricker, and Muhammad Zeshan Afzal (2026). *NURBGen: High-Fidelity Text-to-CAD Generation through LLM-Driven NURBS Modeling*. **AAAI 2026** [[Paper](#)]
 - First LLM-driven NURBS-based Text-to-CAD framework.
 - Introduced *PartABC*, a 300K multi-modal CAD dataset with NURBS and captions.
 - Proposed hybrid analytic-NURBS representation for robust CAD generation.
 - Achieved state-of-the-art fidelity on CAD generation benchmarks.
- **MARVEL-40M+**: Sankalp Sinha*, Mohammad Sadil Khan*, **Muhammad Usama**, Shino Sam, Didier Stricker, Sk Aziz Ali, and Muhammad Zeshan Afzal. (2025). *MARVEL-40M+: Multi-Level Visual Elaboration for High-Fidelity Text-to-3D Content Creation*. **CVPR 2025** [[Paper](#)]
 - Introduced the largest 3D captioning dataset (40M+ annotations, 8.9M assets).
 - Proposed multi-level annotation structure for fine-to-coarse 3D understanding.
 - Integrated domain-specific metadata to reduce VLM hallucinations.
 - Developed MARVEL-FX3D, a two-stage text-to-3D generation pipeline.
- **DreamCAD**: Mohammad Sadil Khan, **Muhammad Usama**, Rolandos Alexandros Potamias, Didier Stricker, Muhammad Zeshan Afzal, Jiankang Deng, Ismail Elezi (2026). *DreamCAD: Scaling Multi-modal CAD Generation using Differentiable Parametric Surfaces*. **CVPR 2026** [[Submitted](#)]

EXPERIENCE

- **Research Assistant** Kaiserslautern, Germany
German Research Center (DFKI) Nov 2023 – Present
 - **Automated captioning pipeline** : Built an end-to-end pipeline to auto-caption 1M 3D assets using vision-language models (VLMs); orchestrated multi-view renders and metadata extraction with Blender.
 - **Fine-tuning LLMs** : Researched and implemented fine-tuning strategies for large language models (LLMs), including Qwen-3B and Qwen-7B, using parameter-efficient methods such as LoRA.
 - **Long Context Fine-tuning** : Explored techniques for long-context fine-tuning of LLMs, evaluating approaches like LongLoRA and other context-extension strategies to enhance model performance on long-sequence tasks.
 - **Ethical filtering** : Designed an LLM-based ethical filtering layer with prompt engineering to detect, redact, and rewrite problematic captions; reduced ethical concerns by 80 percent relative to the unfiltered output.
 - **Pose-guided image generation**: Gained in-depth knowledge of diffusion models through extensive research, evaluated existing methodologies and continuously analyzed ongoing advancements in pose-guided image generation.
 - **Visual Machine Learning**: Developed backend services for a visual machine learning platform that enables users to create, train, and deploy machine learning models through an intuitive user interface. The system supports training on both the DFKI cluster and on-premises MindGarage CPU nodes, providing flexible computational resources for various project needs.
 - **Parallel Distributed Training**: Developed highly efficient software to run machine learning models distributed over clusters, which significantly optimized computational performance.

- **Software Engineer Machine Learning Infrastructure**

Redmond, USA (remote)

Retrocausal

Jul 2022 – Oct 2023

- **Software Packaging:** Led the software packaging of our system for client deployment, tailoring it to diverse Retrocausal clients' factory environments. This contributed to expanding our client base and securing 2nd funding rounds from venture capitals in late 2023. It's a proud milestone in my career, showcasing global recognition and the widespread use of my work.
- **System Scalability:** Successfully enhanced system scalability and efficiency by eliminating the practice of pooling for continuous API calls. I implemented Redis and SQS to facilitate message transmission, ensuring seamless data loading on demand. This strategic integration significantly reduced API calls during both long-running tasks and continuous loading operations, thereby optimizing overall system efficiency.
- **Labelling Tool:** Built an efficient object labelling method for the system, enabling accurate labelling of 2D static and 3D objects in depth and RGB frames. Designed an intuitive UI, resulting in time savings and enhanced product appeal.
- **On-Premises Architecture Design and Development:** Transformed the entire architecture from cloud to on-premises, minimizing performance loss during migration. Successfully transferred AWS services to Localstack and incorporated alternative solutions for Firebase and other cloud services. Developed a packaging script for a seamless system setup on clients' PCs.
- **Thin Web Client :** Developed a lightweight web client version to replace the bulky desktop application UI. Implemented WebRTC for seamless streaming and data channel communication.

- **Software Engineer**

Karachi, Pakistan

Aga Khan Development Network Digital Health Resource Centre (AKDN)

Jul 2020 – Mar 2022

- **Full-Stack Development:** Worked on scalable, multi-module healthcare system integrating patient records, doctor scheduling, appointments, billing, inventory, and payment processing for the largest hospital network in Karachi and its Nairobi subsidiaries. Utilized .NET, NestJS, and Angular for full-stack implementation.
- **Deployment Automation and Containerization:** Led the initiative to modernize deployment workflows by introducing Docker and Docker Compose for standardized environments, and GitHub Actions for automated CI/CD eliminating manual deployment steps and significantly reducing production errors.
- **Analytics Automation:** Wrote Firebase functions to generate daily analytics reports (cron) using Google BigQuery and distributed them automatically through email.

EDUCATION

- **Technische Universität Kaiserslautern-Landau (RPTU)**

Kaiserslautern, Germany

Master of Science in Computer Science

Oct 2023 – Present

- **Master's Thesis** (in progress): Exploring novel approaches in multimodal 3D Retrieval Framework (targeting submission to ECCV 2026)

- **National University of Computer and Emerging Sciences (NUCES-FAST)**

Karachi, Pakistan

Bachelor of Science in Computer Science

Aug 2018 – Jul 2022

SKILLS

- **Programming Languages:** Python, C/C++, Java, JavaScript, C#, Dart
- **ML Architecture:** VLM-LLM Inference, Transformers, Diffusion Model, Stable Diffusion, LORA
- **Software Engineering Frameworks :** .NET Core, Django, Spring Boot, ExpressJs, NestJs, Flutter, Java Android Development
- **ML Libraries :** PyTorch, PyTorch Lightning, Python OCC, vLLM, Blender Python,
- **Interpersonal :** Creative Thinker, Problem solver, Collaborative Engineer and Researcher, Quick and Adaptive Learner